

SQL String Functions

07 · String Functions — Complete Reference Guide

Table of Contents

1. CONCAT — Concatenation
2. LOWER / UPPER — Case Transformation
3. TRIM — Remove White Spaces
4. REPLACE — Replace / Remove Values
5. LEN — String Length
6. LEFT / RIGHT — Substring Extraction
7. SUBSTRING — Extract Substrings
8. Nesting Functions

1. CONCAT() — String Concatenation

→ *Concatenate first name and country into one column*

```
SELECT
    CONCAT(first_name, '-', country) AS full_info
FROM customers
```

2. LOWER() & UPPER() — Case Transformation

→ *Convert the first name to lowercase*

```
SELECT
    LOWER(first_name) AS lower_case_name
FROM customers
```

→ *Convert the first name to uppercase*

```
SELECT
    UPPER(first_name) AS upper_case_name
FROM customers
```

3. TRIM() — Remove Leading & Trailing Spaces

→ *Find customers whose first name contains leading or trailing spaces*

```
SELECT
    first_name,
    LEN(first_name)          AS len_name,
    LEN(TRIM(first_name))    AS len_trim_name,
    LEN(first_name) - LEN(TRIM(first_name)) AS flag
FROM customers
WHERE LEN(first_name) != LEN(TRIM(first_name))
```

4. REPLACE() — Replace or Remove a Value

→ *Remove dashes from a phone number / replace with slash*

```
SELECT
    '123-456-7890'          AS phone,
    REPLACE('123-456-7890', '-', '/') AS clean_phone
```

→ *Replace file extension from .txt to .csv*

```
SELECT
    'report.txt'          AS old_filename,
    REPLACE('report.txt', '.txt', '.csv') AS new_filename
```

5. LEN() — String Length

→ *Calculate the length of each customer's first name*

```
SELECT
    first_name,
    LEN(first_name) AS name_length
FROM customers
```

6. LEFT() & RIGHT() — Substring Extraction

→ *Retrieve the first 2 characters of each first name*

```
SELECT
    first_name,
    LEFT(TRIM(first_name), 2) AS first_2_chars
FROM customers
```

→ *Retrieve the last 2 characters of each first name*

```
SELECT
    first_name,
    RIGHT(first_name, 2) AS last_2_chars
```

FROM customers

7. SUBSTRING() — Extracting Substrings

→ *Remove the first character from each customer's first name*

```
SELECT
    first_name,
    SUBSTRING(TRIM(first_name), 2, LEN(first_name)) AS trimmed_name
FROM customers
```

8. Nesting Functions

→ *Nest LOWER inside UPPER — result is always uppercase*

```
SELECT
    first_name,
    UPPER(LOWER(first_name)) AS nesting
FROM customers
```